Supplementary Material
Naïve Difficulty Conditioning Fails on ShapeNet-55: A Matched-Control Audit

Wenbo Gong *spacetime@tju.edu.cn*
School of Electrical and Information Engineering
Tianjin University, Tianjin, China

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This supplementary collects full-precision numerical artefacts and auditing scripts that support the main paper (§§5.2, 5.3, 5.5 and 6). Each diagnostic section below points to the JSON artefact and Python script behind it; main paper Table 1 lists which results are backed by released artefacts and which survive only as recorded values.

1 Test-time Ensembling Full Results (main §6.4)

Subset definition. For efficiency, the Decision-time Routing (DTR) probe evaluates every ensemble strategy on a deterministic prefix of the official ShapeNet-55 Hard split: the *first* 100 *test samples* (in taxonomy-sorted order, identical across strategies) and the *first* 2 *of* 8 *evaluation viewpoints per sample*, for a total of $N = 200$ completion queries spanning 18 taxonomy categories (the `n_total_samples` field of both JSON artefacts counts the 100 samples; the viewpoint count is recorded in their notes). The first `consumed_samples` list is archived verbatim in `vanilla_ft_subset_hard_100x2.json` so that any re-run can be replayed on the exact same deterministic indices.

Strategies evaluated. Let $d \in \{S, M, H\}$ denote the three difficulty modes passed to DiffCond at test time and $\hat{d}$ the predictor output.

- **cond_easy / cond_median / cond_hard**: feed the fixed label d regardless of sample.
- **predict**: feed $\hat{d}$ (argmax of the CE head).
- **mean**: average the three $\mathbf{q}'$-injected predictions.
- **weighted**: predictor-probability-weighted average of the three predictions.
- **oracle_best_cdl2**: per-sample pick the condition whose completion scores lowest CD-ℓ_2 against ground-truth (optimal routing upper bound; not practical).

Vanilla-FT is re-evaluated on the same prefix subset at ep-30 as a same-subset apples-to-apples comparator.

Key readings. (i) Non-oracle CD-ℓ_2 ranges 1.662–1.696, i.e., within 1.4% of `cond_hard`—closely clustered, consistent with the training-time dead ranking-delta branch. (ii) The per-sample `oracle_best_cdl2` at 1.571 is the theoretical upper bound of any discrete-conditioning routing strategy; at 9.81% worse than Vanilla-FT on the identical subset, it shows that no routing over the three trained conditionings can recover the matched-vanilla representation quality. (iii) Predictor-weighted averaging is a no-op: it differs from `cond_hard` by < 0.001 CD-ℓ_2, because the predictor output is near-identical to `cond_hard` on this subset (the Hard bin).

Provenance. Raw artefacts: `data/results/dtr_results.json` (7 strategies), `data/results/vanilla_ft_subset_hard_100x2.json` (matched-vanilla on the same subset, with full `consumed_samples` list). Scripts: `tools/eval_dtr.py`, `tools/eval_vanilla_subset.py`. Checkpoints: the matched DiffCond and Vanilla-FT (seed 42) final fine-tuned checkpoints from the 30-epoch trajectory.

Table 1: DTR full results on the Hard-bin subset (100 samples × 2 viewpoints = 200 evaluations, 18 categories). Metrics: F-Score@1%, CD-ℓ_1 $\times 10^3$, CD-ℓ_2 $\times 10^3$, EMD $\times 10^3$ (lower is better except F-Score). Vanilla-FT is evaluated on the identical subset for apples-to-apples comparison; best CD-ℓ_2 in bold. $\Delta_{\text{Vanilla-FT}}$ is the relative change from Vanilla-FT on CD-ℓ_2 (negative means DiffCond strategy worse), computed from full-precision values, so a back-computation from the rounded entries can differ by up to 0.05pp. All numbers computed by `tools/eval_dtr.py` (for DiffCond strategies) and `tools/eval_vanilla_subset.py` (for Vanilla-FT).

Strategy	F-Score ↑	CD-ℓ_1 ↓	CD-ℓ_2 ↓	EMD ↓	$\Delta_{\text{Vanilla-FT}}$
Vanilla-FT (ep-30, same subset)	0.313	17.06	**1.430**	35.13	—
cond_easy	0.305	18.07	1.696	36.90	−18.56%
cond_median	0.307	17.85	1.667	35.96	−16.52%
cond_hard	0.309	17.65	1.673	34.23	−16.95%
predict ($\hat{d}$)	0.309	17.65	1.673	34.31	−16.97%
mean	0.307	17.84	1.662	35.77	−16.20%
weighted	0.309	17.65	1.672	34.49	−16.92%
oracle_best_cdl2 (upper bound)	0.308	17.52	1.571	34.97	−9.81%

2 Difficulty Predictor Calibration (main §6.1, Finding 2)

The auxiliary cross-entropy head predicts difficulty from the encoder global feature $\mathbf{z}$. We probe predictor quality over $N = 12{,}000$ held-out (viewpoint, difficulty)-labelled evaluations sampled from the official S/M/H eval, balanced at 4,000 per oracle label {Simple=0, Moderate=1, Hard=2}.

Aggregate metrics. $N = 12{,}000$; accuracy 0.99875; Pearson correlation predicted probability vs. one-hot oracle 0.9991; Brier score 0.0021; expected calibration error (ECE, 10-bin) 0.0051. The predictor is near-perfect on this probe and exhibits low calibration error.

Table 2: Predictor confusion matrix (rows: oracle label; columns: predicted label). Diagonal mass reveals the near-identity of predictor output to oracle.

Oracle \ Predicted	Simple (0)	Moderate (1)	Hard (2)	Row total
Simple (0)	**3997**	3	0	4000
Moderate (1)	3	**3991**	6	4000
Hard (2)	0	3	**3997**	4000
Column total	4000	3997	4003	12,000

Table 3: Per-class predictor performance.

Class	Precision	Recall	F1	Support
Simple (0)	0.99925	0.99925	0.99925	4000
Moderate (1)	0.99850	0.99775	0.99812	4000
Hard (2)	0.99850	0.99925	0.99888	4000

Key readings. (i) The predictor is empirically identical to the oracle on 11,985/12,000 samples (99.88%), and the Oracle vs. Predict inference modes in Table 2 of the main paper differ by ≤ 0.004 CD-ℓ_2 per bin across all $3 \times 3 = 9$ seed×bin pairs (Table 5; max deviation 0.004 at seed 43, Hard; other 8 pairs ≤ 0.002). (ii) Low Brier and ECE indicate the softmax output is well-calibrated, not just argmax-correct.

Provenance. Raw artefact: `data/results/diffcond_predictor_calibration.json`. Script: `tools/eval_predictor_calibration.py`. Checkpoint: the DiffCond final fine-tuned checkpoint (seed 42).

Table 4: Per-oracle mean predicted probability mass across the three classes (should concentrate on the diagonal).

Oracle label	$\bar{p}$(Simple)	$\bar{p}$(Moderate)	$\bar{p}$(Hard)
Simple	**0.9973**	0.0024	0.0003
Moderate	0.0051	**0.9919**	0.0031
Hard	0.0002	0.0081	**0.9917**

Table 5: Per-(seed, bin) |Predict − Oracle| in CD-ℓ_2×10^3 for DiffCond at ep-30. All 9 pairs within 0.004 (max at seed 43, Hard). Seed 42 evaluated together with the main matched-control run; seeds 43, 44 evaluated in a subsequent replication batch on identical checkpoints.

Seed	Simple	Moderate	Hard
42	0.000	0.002	0.001
43	0.000	0.001	**0.004**
44	0.000	0.001	0.000

3 Static Weight Audit (main §6.1, Findings 1–3)

We directly inspect the parameters of the trained DiffCond checkpoint and compare to the Vanilla-FT matched vanilla checkpoint (both at ep-30, both warm-started from Baseline).

Table 6: DiffCond-specific parameters in DiffCond at ep-30, full values. Δ_{diff} is the per-query ranking residual (last layer zero-initialised by construction). $\mathbf{W}_{\text{proj}}$ is the decoder-query projection (fully zero-initialised by construction). $\mathbf{W}_g$ is the gated-injection gate weight (trunc-normal initialised).

Parameter / quantity	Shape / detail	Value
Δ_{diff} last-layer weight norm	[1, 128]	**0.000**
Δ_{diff} last-layer weight abs-max		**0.000**
Δ_{diff} last-layer bias norm	[1]	**0.000**
$\mathbf{W}_{\text{proj}}$ weight Frobenius norm	[384, 384]	4.610
$\mathbf{W}_{\text{proj}}$ weight ℓ_∞		0.0649
$\mathbf{W}_{\text{proj}}$ bias norm	[384]	0.1393
$\mathbf{W}_g$ weight Frobenius norm		11.814
$\mathbf{W}_g$ weight std		0.0218
$\mathbf{W}_g$ bias norm		0.4052
$\mathbf{W}_g$ bias mean		−0.0056
Difficulty Embedding $\mathbf{E}$ shape		[3, 384]
$\|\mathbf{E}_{\text{Simple}}\|_2$		0.6669
$\|\mathbf{E}_{\text{Moderate}}\|_2$		0.2086
$\|\mathbf{E}_{\text{Hard}}\|_2$		0.9794
$\cos(\mathbf{E}_{\text{Simple}}, \mathbf{E}_{\text{Moderate}})$		+0.5714
$\cos(\mathbf{E}_{\text{Simple}}, \mathbf{E}_{\text{Hard}})$		−0.9539
$\cos(\mathbf{E}_{\text{Moderate}}, \mathbf{E}_{\text{Hard}})$		−0.7416
Auxiliary classifier last-layer weight norm		1.101

Finding 1 support. The Δ_{diff} last-layer weight and bias are *identically zero* (not small, not near-zero). This is confirmed architecturally: the downstream `argsort+gather` operation is non-differentiable with respect to the pre-selection scores, so the reconstruction loss back-propagates no gradient to Δ_{diff}. Initialised at zero and receiving no gradient for 30 epochs, the branch cannot escape zero; this is the *architecturally dead* property cited in the main text.

Finding 2 support. The zero-initialised projection $\mathbf{W}_{\text{proj}}$ is non-zero at ep-30 (norm 4.61, ℓ_∞ 0.065), and so is the zero-initialised gate bias (norm 0.41, mean -0.0056). The gate weight $\mathbf{W}_g$ is drawn from a trunc-normal initialiser with standard deviation 0.02 and has final element standard deviation 0.0218 (Frobenius norm 11.81); its initial tensor was not archived, so its change is not measured. These statistics show that the two zero-initialised parameters changed during training; none of them measures the injection path's contribution to the output, and we make no claim about its magnitude. The three embeddings $\mathbf{E}_d$ are geometrically distinct (Simple-Hard cosine -0.95; Simple-Moderate $+0.57$), so the injection path receives distinct inputs for the three difficulty levels. (At epoch 0 the projection is zero by construction, which is the property the epoch-0 probe of main-paper §6.2 relies on.)

Table 7: Shared-backbone drift relative to Baseline at ep-30 (Frobenius distance normalised by the corresponding Baseline parameter norm). All parameters present in both Baseline and the fine-tuned checkpoint are compared; Δ_{pp} is DiffCond minus Vanilla-FT (positive means DiffCond drifted further). *base_model core drift* (encoder + decoder transformer trunk) is the key reading for main-paper §6.1, Finding 3.

Sub-module	DiffCond drift	Vanilla-FT drift	Δ_{pp}	n_{params}
`base_model` (encoder+decoder transformer)	23.81%	21.57%	+2.24	30,063,585
`decode_head`	18.03%	17.91%	+0.11	418,352
`increase_dim`	10.65%	10.65%	0.00	1,447,937
`reduce_map`	44.66%	43.54%	+1.12	542,208
Overall (32.5M shared params)	10.65%	10.65%	0.00	32,472,082

Finding 3 support. The overall shared-backbone drift is identical between DiffCond and Vanilla-FT at 10.65%, yet the `base_model` encoder-decoder core alone drifts +2.24pp further in DiffCond. Because the two candidates share warm-start source, schedule, seed, crop path, and reconstruction loss, the +2.24pp excess drift is consistent with the two DiffCond-exclusive gradient sources (the auxiliary CE loss on the encoder global feature and the gated-injection path into the decoder), but a final-parameter distance does not identify the contribution of either source. The three-seed ablation in main-paper §5.3 does not resolve the CE contribution (its CE-removal contrast spans zero widely); this per-module drift table is an observation consistent with, and not evidence for, a CE-driven mechanism.

Provenance. Raw artefact: `data/results/ckpt_static_audit.json`. Script: `tools/ckpt_static_audit.py`. Checkpoints inspected: the DiffCond and matched Vanilla-FT final fine-tuned checkpoints (seed 42), and the pretrained backbone reference checkpoint (for drift computation).

4 Cross-Backbone Replication: Full Per-Seed Table (main §5.5)

Provenance. All values are computed by the aggregate script `tools/aggregate_results.py` from the per-(seed, mode) S/M/H evaluation summary files released in the submission bundle for the UpTrans Vanilla-FT and DiffCond checkpoints at seeds $\{42, 43, 44, 45\}$. The aggregate JSON and paired-statistics TSV are released with the bundle; producing script: `tools/paired_stats.py`. The initial 3-seed S/M/H eval (6 checkpoints $\times$ 3 modes $= 18$ runs) was launched from a single master pipeline; all 9/9 phase transitions were zero-gap and the full pipeline emitted 0 NaN / Error / Traceback events across the entire $\approx$82h wall time (stage master log, retained locally and released in the bundle). A subsequent fourth-seed (`seed 45`) follow-up run added 2 checkpoints (Vanilla, DiffCond) and 6 S/M/H evaluations under the same protocol; we tighten the 95% CI on the focal cell from df $= 2$ to df $= 3$ as a result.

Setup. For both UpTrans and UpTrans_DiffCond candidates, 30-epoch fine-tune from a shared 400-epoch UpTrans warm-start (the pretrained ShapeNet-55 checkpoint released by the ODGNet authors); seeds $\{42, 43, 44, 45\}$; identical 8-view S/M/H evaluation protocol as in main paper §5.1 (Setup). Tables 8 and 9 report per-seed Overall metrics.

Table 8: UpTrans Vanilla-FT per-seed Overall metrics on ShapeNet-55 S/M/H. CD scaled by 10^3.

Seed	Mode	F-Score ↑	CD-ℓ_1 ↓	CD-ℓ_2 ↓
42	Simple	0.456	11.440	0.495
42	Moderate	0.442	13.078	0.738
42	Hard	0.389	16.803	1.504
43	Simple	0.456	11.445	0.488
43	Moderate	0.442	13.079	0.724
43	Hard	0.381	17.309	1.583
44	Simple	0.447	11.480	0.508
44	Moderate	0.438	12.998	0.735
44	Hard	0.389	16.431	1.418
45	Simple	0.445	11.489	0.508
45	Moderate	0.439	13.011	0.734
45	Hard	0.389	16.697	1.470

Table 9: UpTrans-DiffCond per-seed Overall metrics on ShapeNet-55 S/M/H. CD scaled by 10^3.

Seed	Mode	F-Score ↑	CD-ℓ_1 ↓	CD-ℓ_2 ↓
42	Simple	0.457	11.438	0.488
42	Moderate	0.445	13.018	0.721
42	Hard	0.388	16.955	1.499
43	Simple	0.453	11.546	0.496
43	Moderate	0.440	13.116	0.729
43	Hard	0.385	16.934	1.490
44	Simple	0.449	11.427	0.501
44	Moderate	0.440	12.986	0.730
44	Hard	0.390	16.404	1.398
45	Simple	0.443	11.697	0.510
45	Moderate	0.434	13.230	0.741
45	Hard	0.383	16.872	1.477

Paired statistical detail. The full 9-cell paired-seed CI summary (3 modes × 3 metrics) is released with the submission bundle and is reproducible via `python tools/paired_stats.py`. The headline summary for CD-ℓ_2 matches main paper Table 4 (the cross-backbone CD-ℓ_2 paired analysis); F-Score and CD-ℓ_1 paired means and 95% CIs (df = 3) are summarised in Table 10.

Reading. Across all 9 (mode, metric) cells, the 95% CI crosses zero (lower bound below 0, upper bound above 0) and the lower bound is strictly below the +3% method-bar threshold. We emphasise that the data are equally consistent with a small genuine improvement (which would require many more seeds to detect) and with no architecture-general effect at all; the central claim of main paper §5.5 is the cautious one—"no large architecture-general gain"—and is robust to either reading.

Stop-rule for the seed 45 follow-up. Seed 45 was added under a one-step stop-rule declared in advance (in the same sense as the other thresholds in main paper §3.4: declared before running the additional seed, but not filed in a public preregistry) and not in response to its own outcome: extend the SN55 cross-backbone seed set from $\{42, 43, 44\}$ to $\{42, 43, 44, 45\}$ exactly once, re-run the aggregate and paired-statistics scripts with the new seed list, and report the n=4 numbers regardless of whether they tighten or loosen the original conclusion. Seed 45 was not chosen on the basis of its metric values (the run was launched before any seed-45 evaluation existed), and no further seeds were added or removed after observing the seed-45 outcome. This single one-step extension is reflected in the n=4 paired analysis above.

Table 10: UpTrans cross-backbone paired-seed analysis, all 9 (mode, metric) cells. Per-seed paired % improvement is sign-aware: for CD-ℓ_1 and CD-ℓ_2 lower is better, for F-Score higher is better. unadjusted 95% CI uses Student's t at df = 3 ($t_{0.975} = 3.18$). "Beats +3%?" tests whether the CI lower bound clears the pre-specified method-bar of main paper §3 (proposed protocol). Numbers are reproduced verbatim from the paired-statistics TSV released with the submission bundle (rounded to 2 decimals).

Mode	Metric	Mean paired %	95% CI	Beats +3%?
Simple	F-Score	−0.11%	[−0.95, +0.73]%	no
Simple	CD-ℓ_1	−0.55%	[−2.16, +1.05]%	no
Simple	CD-ℓ_2	+0.19%	[−2.17, +2.55]%	no
Moderate	F-Score	−0.11%	[−1.45, +1.22]%	no
Moderate	CD-ℓ_1	−0.35%	[−1.84, +1.14]%	no
Moderate	CD-ℓ_2	+0.33%	[−2.04, +2.71]%	no
Hard	F-Score	−0.12%	[−1.85, +1.61]%	no
Hard	CD-ℓ_1	+0.09%	[−2.27, +2.46]%	no
Hard	CD-ℓ_2	+1.79%	[−2.72, +6.29]%	no

Multiplicity caution. Table 10 reports nine unadjusted 95% CIs (3 modes × 3 metrics). They are not independent: all nine come from the same four seed-matched model pairs, and no multiplicity adjustment is applied. We do not interpret any single nominally-significant cell as evidence of an effect; in this work all nine cells happen to span zero, so multiplicity does not change the qualitative reading. We flag this explicitly because a hostile reading could otherwise conflate "1 of 9 cells closer to the +3% bar" with "cross-backbone DiffCond beats the +3% bar"—which it does not, on this evidence.

Holm–Bonferroni sensitivity (paper-wide). We did not pre-register a familywise multiplicity adjustment, but for transparency we report a post-hoc check. The family covers the main matched-control comparison (main paper Table 2: four paired-Δ t-tests across Simple, Moderate, Hard, and Mean for AdaPoinTr) and the cross-backbone table above (9 CIs), i.e. 13 in-scope t-tests at nominal $\alpha = 0.05$. Recomputing the one-sample paired t statistics from the released per-seed tables, the two smallest p-values are Mean CD-ℓ_2 ($p \approx 0.0046$) and Hard CD-ℓ_2 ($p \approx 0.0087$). Under a Holm–Bonferroni adjustment, the strictest threshold for the smallest p-value is $\alpha/13 \approx 0.0038$; the second is $\alpha/12 \approx 0.0042$. *No contrast in the* 13*-test family formally clears the Holm-adjusted threshold* (the main Mean and Hard contrasts are the closest, but both fall just above their respective thresholds). We therefore report all paired CIs as unadjusted descriptive evidence, not as familywise-significance claims. We retain the sign-consistency observation (six negative paired-Δ entries across three bins × two inference settings) only as a descriptive structural note, *not* as six independent significance tests: Oracle and Predict share the same trained checkpoint and the predictor is near-identical to oracle (Predict tracks Oracle within 0.004 CD-ℓ_2 on every per-(seed, bin) cell of Table 5), so the two inference settings produce highly correlated readouts rather than independent replicates. Our headline reading therefore rests on the matched-control point estimate (main paper Table 2) and its unadjusted CIs, deliberately framed as descriptive rather than inferential.

5 PCN Sanity Check: Per-Seed Table (main §5.5)

Provenance and caveats. The PCN cross-benchmark sanity check uses a 2-seed UpTrans Vanilla / UpTrans-DiffCond pair on the official PCN test split (1,200 samples across 8 categories). PCN does not provide native S/M/H difficulty labels; we compute a nearest-neighbour difficulty proxy once on the training set and archive the resulting JSON in `data/PCN/difficulty_proxy.json`. We position this as a supplementary sanity check rather than core cross-backbone evidence because: (i) the difficulty proxy has no native ground-truth calibration, (ii) only 2 seeds were run (vs. 4 for the SN55 cross-backbone replication), and (iii) PCN's category distribution and crop pipeline differ substantially from ShapeNet-55, so any PCN result is at most a coarse generalisation probe.

Table 11: PCN cross-backbone sanity check, per-seed Overall metrics on the official PCN test split. CD scaled by 10^3. $\Delta\%$ is DiffCond *minus* Vanilla-FT, sign-aware (for CD lower is better; for F-Score higher is better). Mean rows and $\Delta\%$ are computed from full-precision per-seed values; small differences between the displayed $\Delta\%$ and a back-computation from the rounded mean entries are expected (cf. main paper Table 2). All mean deltas are at most 1.1% and below the +3% method-bar, and on every metric the two seeds' deltas have opposite signs.

Family	Seed	F-Score ↑	CD-ℓ_1 ↓	CD-ℓ_2 ↓
Vanilla-FT	42	0.828	6.645	0.214
Vanilla-FT	43	0.826	6.622	0.208
Vanilla-FT	mean	0.827	6.634	0.211
DiffCond	42	0.827	6.605	0.201
DiffCond	43	0.828	6.624	0.216
DiffCond	mean	0.828	6.615	0.209
$\Delta\%$ (mean)	—	+0.06%	+0.3%	+1.1%

Reading. The 2-seed mean deltas are at most $\sim 1.1\%$ on every metric and all fall well below the pre-specified +3% method-bar. A paired 95% CI is not meaningful at $n = 2$ (df = 1, $t_{0.975} = 12.71$ inflates the CI half-width past any informative bound), so we report the deltas descriptively and do not claim either a positive or a negative effect from the 2 PCN seeds. The data are consistent with the SN55 cross-backbone reading (no large architecture-general gain) and inconsistent with the AdaPoinTr double-digit inflation pattern of main paper §3.2.

6 AdaPoinTr Matched-Control Full Per-Seed Metrics (main §5.2)

Why this section. The proposed protocol (main paper §3) calls for reporting F-Score@1%, CD-ℓ_1, and CD-ℓ_2 for every candidate × bin pair. Main paper Table 2 compresses the headline matched-control comparison to CD-ℓ_2 for readability. This section provides the full F-Score / CD-ℓ_1 / CD-ℓ_2 table per seed for all three matched-control candidates (Vanilla-FT, DiffCond Oracle, DiffCond Predict) on the official 8-viewpoint S/M/H protocol.

Reading. Predict tracks Oracle within 0.005 on F-Score, 0.011 on CD-ℓ_1, and 0.004 on CD-ℓ_2 across all 9 seed×bin pairs (the near-perfect predictor calibration of Table 2 explains why). DiffCond Oracle/Predict regress against Vanilla-FT on every cell of the matched-control protocol, consistent with the 3-seed CD-ℓ_2 aggregate in main paper Table 2.

Power and minimum detectable effect at $n = 3$. The AdaPoinTr matched-control comparison uses $n = 3$ seeds. At df = 2 and $t_{0.975} = 4.30$, the minimum detectable Cohen d at 80% power for a two-sided paired-t test is approximately $d \approx 3.3$. Mapped through the observed across-seed paired-Δ SDs in our data—Hard CD-ℓ_2 SD ≈ 0.89pp (tight) and Hard-Specialist CD-ℓ_2 SD ≈ 3.82pp (wide)—this corresponds to absolute minimum detectable effects of roughly ±3pp in the tight-SD regime and roughly ±12.5pp in the wide-SD regime. Paired-Δ effects below these magnitudes cannot be reliably detected at our seed budget, and we therefore treat marginal CIs (notably the Hard-Specialist mean $+7.38\% \pm 3.82$pp, whose CI crosses zero) as descriptive of the 3-seed estimate rather than as evidence of magnitude or absence of effect.

7 Three-Seed Ablation: Per-Seed Values

Main-paper Table 3 reports the four-condition ablation as paired per-seed deltas with Student-t intervals. This table lists the underlying per-seed CD-ℓ_2 values on the official S/M/H protocol. Seeds 43 and 44 for v1-λ0 and v2b were added after review under the identical matched-control recipe (Baseline warm-start, 30 epochs, DiffCond crop path); the seed-42 rows are the values of the submitted single-seed matrix. ΔMean is the paired improvement (%) of the bin-averaged CD over the Vanilla-FT run of the same seed (negative

Table 12: Matched-control full per-seed metrics on the official ShapeNet-55 8-viewpoint S/M/H protocol. CD scaled by 10^3. Vanilla-FT, DiffCond Oracle, and DiffCond Predict share warm-start source, 30-epoch schedule, seed set $\{42, 43, 44\}$, and crop pipeline. Numbers from `data/results/all_results.json` in the released codebase; reproducible via the released eval scripts.

Family	Seed	Mode	F-Score ↑	CD-ℓ_1 ↓	CD-ℓ_2 ↓
Vanilla-FT	42	Simple	0.398	12.431	0.568
Vanilla-FT	42	Moderate	0.380	13.690	0.810
Vanilla-FT	42	Hard	0.337	16.905	1.529
Vanilla-FT	43	Simple	0.399	12.477	0.582
Vanilla-FT	43	Moderate	0.380	13.765	0.838
Vanilla-FT	43	Hard	0.334	17.104	1.598
Vanilla-FT	44	Simple	0.398	12.394	0.572
Vanilla-FT	44	Moderate	0.381	13.610	0.813
Vanilla-FT	44	Hard	0.339	16.777	1.529
DiffCond Oracle	42	Simple	0.388	12.666	0.601
DiffCond Oracle	42	Moderate	0.375	13.829	0.836
DiffCond Oracle	42	Hard	0.327	17.270	1.619
DiffCond Oracle	43	Simple	0.392	12.683	0.606
DiffCond Oracle	43	Moderate	0.376	13.889	0.853
DiffCond Oracle	43	Hard	0.328	17.520	1.695
DiffCond Oracle	44	Simple	0.394	12.641	0.587
DiffCond Oracle	44	Moderate	0.378	13.933	0.848
DiffCond Oracle	44	Hard	0.335	17.160	1.597
DiffCond Predict	42	Simple	0.388	12.667	0.601
DiffCond Predict	42	Moderate	0.375	13.834	0.838
DiffCond Predict	42	Hard	0.327	17.273	1.620
DiffCond Predict	43	Simple	0.392	12.684	0.606
DiffCond Predict	43	Moderate	0.376	13.891	0.854
DiffCond Predict	43	Hard	0.328	17.530	1.699
DiffCond Predict	44	Simple	0.394	12.642	0.587
DiffCond Predict	44	Moderate	0.378	13.936	0.849
DiffCond Predict	44	Hard	0.335	17.161	1.597

= worse). Numbers from `data/results/gap_a_results.json`, regenerated by `tools/gap_a_results.py` from the per-seed evaluation logs.

8 Retained Checkpoints: SHA-256 Manifest

Table 14 lists every AdaPoinTr-family checkpoint retained by the authors, with its role in the paper, its seed and epoch, and its SHA-256 digest. The manifest was computed on the local copies on 2026-08-21 (22 files) and extended on 2026-09-18 with the four added-seed ablation checkpoints, whose digests equal the values sealed in the provenance record of the added-seed run (`data/results/gap_a_provenance/`). Three retained checkpoints are not used for any reported number and are marked as non-paper. These checkpoints are released with the hashes below at `https://huggingface.co/spacetime1008/diffcond-matched-control-audit`. The UpTrans and PCN checkpoint binaries were not retained after the remote instance was released and therefore do not appear here.

9 Retained Training and Evaluation Logs: Inventory

The training and evaluation logs behind the reported numbers are retained on the authors' local archive; Table 15 inventories them as of 2026-09-18. For the UpTrans and PCN runs every training and evaluation log is retained as a per-run file; for the AdaPoinTr-family runs the logs are retained either as per-run files or inside the batch logs that carry the per-epoch training output and the evaluation results of a whole batch of

Table 13: Per-seed CD-ℓ_2 $\times 10^3$ for the four ablation conditions (seeds 42, 43, 44). The per-seed ΔMean spread is 1.1pp for DiffCond (full), 11.3pp for v1-λ0 and 6.0pp for v2b.

Candidate	Seed	Simple	Moderate	Hard	Mean CD	ΔMean (%)
Vanilla-FT	42	0.568	0.810	1.529	0.969	0.00
Vanilla-FT	43	0.582	0.838	1.598	1.006	0.00
Vanilla-FT	44	0.572	0.813	1.529	0.971	0.00
DiffCond (full)	42	0.601	0.836	1.619	1.019	−5.13
DiffCond (full)	43	0.606	0.853	1.695	1.051	−4.51
DiffCond (full)	44	0.587	0.848	1.597	1.011	−4.05
v1-λ0 (CE off, rank on)	42	0.573	0.823	1.580	0.992	−2.37
v1-λ0 (CE off, rank on)	43	0.585	0.834	1.547	0.989	+1.72
v1-λ0 (CE off, rank on)	44	0.604	0.877	1.713	1.065	−9.61
v2b (CE off, rank off)	42	0.596	0.854	1.550	1.000	−3.20
v2b (CE off, rank off)	43	0.586	0.835	1.603	1.008	−0.20
v2b (CE off, rank off)	44	0.554	0.794	1.485	0.944	+2.78

Table 14: SHA-256 manifest of the 26 retained AdaPoinTr-family checkpoints (ShapeNet-55). "Epoch": fine-tuning epoch of the saved weights; "best" and "last" are the training framework's best-validation and final checkpoints of the same run.

Paper role	Seed	Epoch	SHA-256
Baseline: 300-epoch from-scratch AdaPoinTr (warm-start source of every fine-tune)	–	best	f1acb03baa1f80c5c4eeb361d9b23a0193e9891d31df1a0de0b4de37bee534b3
Vanilla-FT (matched control)	42	30	afc345f22855db2d8fa233643a1467b3e10edf1dedbbdc9112a5df98d6cd6c9f
Vanilla-FT (matched control)	43	30	8ce7a0ec2158ca5164fbe7ddf7ab2f700af934f1ea244009d114b744bf3ce320
Vanilla-FT (matched control)	44	30	78e8859a985f111837343f58bcfc3ba8b01671707c4b66dc9b820f920c92818c
DiffCond (full)	42	30	05f6840d8a9df127b9cf1d36915f1bd24cc0ca2e492e2a4e50973b5ea048bf28
DiffCond (full)	43	30	84f6e0672ba007de193f6aab5623a2701e7d4b372d08cccdaab20b347237027e
DiffCond (full)	44	30	38f811730d99d352b674ec9498ac7677796c855d259a536fcd3900eb061cbf9a
v1-λ0 (CE off, rank on)	42	30	4b1d63bf8142b713aa9da7e6e21443cae6ae4464af1986c71557a0055d797256
v1-λ0 (CE off, rank on); added after review	43	30	e20a5a987a24575d80383fec7295019d7b281fe1c0b6348ef930384e8178a7e4
v1-λ0 (CE off, rank on); added after review	44	30	1bb39ab2896eea0e926490b660fa4b03f2fde7fd548600b0154db5b69dc4f5a2
v2b (CE off, rank off)	42	30	3f8c329a186a1f204eeefe0027cce0ee6eb250db68146c5657684bd5725b6d28
v2b (CE off, rank off); added after review	43	30	8af6dc2c56f1a0599d8a4b3cb53fa54ce6e3d164636309fda58a0a319605dd28
v2b (CE off, rank off); added after review	44	30	a132394ed4447c4a0b83a77d7c96f6350ca8e1f63ac6797b57710ba89ad8f62c
Hard-Specialist	42	30	4ccecd292fd65559db182902dd617e4a9b651ae6a89cafe5900a2983b8ce8b17
Hard-Specialist	43	30	54be5831d4e9285e67b942df35419385f0032b8e1e9f28f5d32e3ba5447a0d8b
Hard-Specialist	44	30	90f6ae8a17c1f06cdcaef42aef606e47441b7418f67b240d892df97d4eaa2e1f
EasyWide-Specialist	42	30	09d9f3393f9b43a0029dfdc813ed73e8185e14ab648e24f9da39c8f796798518
EasyWide-Specialist	43	30	f4451ce14b6ef3c678deebece44ed60ef166fedfe539cf465ef2258beee627c9
EasyWide-Specialist	44	30	7f9e2b226e35a7b11a1ee0ab0607e42ed74a80c7350bb1c3006d5dcfbf022677
From-scratch 30-epoch baseline rerun (Pitfall 2 audit)	42	best	4698e6dc6bedb0122f2d084db5719e0d3d9b1a063ec2a067e09ff386328b6e6c
From-scratch 30-epoch baseline rerun (Pitfall 2 audit)	42	29	e280aba3ff3a69cfcd2a93a236df0edda13b26cc688114a8720ac9a495d598e0
From-scratch 30-epoch baseline rerun (Pitfall 2 audit)	42	30	ad7cf6f496d0d249c794ffaa17e701ec12b105139483f9e868b2a9fc976d9315
From-scratch 30-epoch baseline rerun (Pitfall 2 audit)	42	last	ae0c26d60deb82a959e13c63aee47916da250e40d1d777c6abbfadbed12995d9
Non-paper: early vanilla warm-start fine-tune with the stock per-sample crop path (development-stage comparator behind main §3.3; superseded by Vanilla-FT; no reported number depends on it)	–	30	41f8468cf9b99801629e93de36b60ee2b64ed1e24cd3a780c2c9419b090802df
Non-paper pilot: Easy-Specialist (not used in the paper)	–	30	3929ef449ed9403ad5d49baf3e37f554b10072799b1217e84e78403839a3c2ac
Non-paper pilot: Median-Specialist (not used in the paper)	–	30	b5e0e48a79e5998ccccbc51d94962d5ffd0022fe9c93973fe66eaf4faab16313

runs. Every experiment log listed here (212 of the 223 files; the other 11 are LaTeX compile logs) is released in the bundle under `logs/`; `logs/README.md` maps the rows of the table to folders and `logs/REDACTIONS.md` lists the four lines from which a tool name was removed.

Table 15: Inventory of the retained `.log` files (local archive, counted 2026-09-18; 223 files in total, of which 11 are LaTeX compile logs of paper drafts and are not experiment logs). Four PCN training logs also exist as copies in the experiments tree, and the 24 UpTrans per-seed, per-bin evaluation summaries derived from the UpTrans evaluation logs are in the bundle.

Content	Files	Status
AdaPoinTr training logs for Baseline, Vanilla-FT and DiffCond at seed 42 and for two non-paper pilots; batch logs with the per-epoch training and evaluation output of the seed-43/44 Vanilla-FT and DiffCond runs, the specialist runs and the seed-42 ablation runs; official S/M/H evaluation logs; epoch-0 and routing sanity logs	16	retained locally
Per-run logs: DiffCond seed-43/44 evaluations (7), Baseline training (1), Vanilla-FT seed-42 training (1), EasyWide-Specialist seed-43/44 training (2) and evaluations (6)	17	retained locally
Orchestration and smoke-test logs, including the Vanilla-FT Hard-subset evaluation and the predictor-calibration run	9	retained locally
UpTrans training logs (Vanilla and DiffCond, seeds 42–45), PCN training logs (2 models × 2 seeds) and the baseline main log	13	retained locally
UpTrans official S/M/H evaluation logs (2 models × 4 seeds × 3 bins)	24	retained locally; summaries in bundle
Development audit of 2026-04-16: epoch-0 checkpoint construction and probe, routing sanity check, batch step logs	7	retained locally
Remote orchestration logs (9) and pre-project pilot logs, non-paper (9)	18	retained locally
From-scratch baseline rerun (Pitfall 2 audit): training (3) and per-bin evaluation (3)	6	retained locally; aggregate JSON in bundle
Added-seed ablation run: training (4), evaluation (12), reproduction gate (3), launcher and dataset-verification logs (39)	58	retained locally; the 4 training and 12 evaluation logs, the sentinels and the parsed metrics are in the bundle
Earlier pull of the same added-seed run (superseded copies)	35	retained locally
Non-paper PCN pilot logs	8	retained locally
UpTrans stage master log	1	retained locally
LaTeX compile logs of paper drafts	11	not experiment logs

Release

The bundle, released at `https://github.com/grenbel/diffcond-matched-control-audit`, ships the YAML configurations, the patched training pipeline (`tools/runner.py`, `tools/builder.py`, `datasets/`), the evaluation / parsing scripts and the audit scripts referenced in §1–6, the aggregate JSON / TSV artefacts behind the numerical results of the main paper and this supplementary (main paper Table 1 marks the single-run probes and development-stage observations that survive only as recorded values), the added-seed ablation artefacts (`data/results/gap_a_results.json`, `tools/gap_a_results.py`, `data/results/gap_a_provenance/`) and the crop-path quantification (`data/results/crop_path_quantification.json`, `tools/crop_path_quantification.py`). *Raw checkpoints are not included in the bundle.* The 26 AdaPoinTr-family checkpoints of Table 14 are retained and released with those hashes on Hugging Face; the UpTrans and PCN checkpoint binaries were not retained after the remote instance was released, so those runs are reproducible only by re-running the released pipeline, with their per-seed evaluation artefacts and configurations in the bundle. The four added-seed checkpoints were sealed and evaluated on a second host of the same GPU model with the same software stack but a different driver and kernel; the main-paper Reproducibility Statement and the provenance README record the details. The experiment logs of Table 15 are in the bundle under `logs/`. The aggregate analyses in this supplementary can be reproduced *deterministically* from the released aggregate artefacts, and the matched-control comparison can be reproduced end-to-end by re-running the released training pipeline from the same upstream PoinTr commit, given the Baseline warm-start checkpoint (released on Hugging Face, or retrained with the upstream configuration).